

Phase 1: DINO v0.0.1 - CPU-Free 8-Bit Logic Sequencer

Objective

Design and build a manual, CPU-free 8-bit sequencer that generates addresses, loads program bytes via DIP switch, and displays stored data sequentially on LEDs.

Component BOM

Qty	Component	Function
2	SN74LS590N	8-bit binary counters (16-bit address generation)
1	MCM60256AP-10	32KB SRAM (program storage)
1	555 Timer	Clock generation (~6.8Hz, 52.7% duty cycle)
1	SPDT switch	LOAD/RUN mode control
2	Pushbuttons	WRITE enable, manual CLOCK
1	8-bit DIP switch	LOAD mode data input
8	LEDs	RUN mode output display
8	10kΩ resistors	Pull-down for DIP switch
8	220-470Ω resistors	LED current limiting
1	74LS645N	Octal bus transceiver (bus arbitration)
1	10kΩ resistor (R1)	555 timer frequency setting
1	100kΩ resistor (R2)	555 timer frequency setting
1	1μF capacitor (C1)	555 timer frequency setting
1	0.1μF capacitor (C2)	555 timer frequency setting
8	0.1μF capacitors	Power supply decoupling
1	4.7μF capacitor	Bulk power supply filtering
-	Breadboards, jumper wires	Construction
1	5V power supply	System power

Bus Architecture

Device	Bus Role	Mode
DIP Switch	Drives bus	LOAD
SRAM	Drives & reads	LOAD & RUN
LED Display	Reads from bus	RUN

74LS645 Control Signals

Mode	/OE	DIR	Function
LOAD	Low	High	DIP → SRAM write
RUN	Low	Low	SRAM → LED read
IDLE	High	Low	Bus tri-stated

Build Steps

1. Program Counter Assembly

- Chain two SN74LS590N counters for 16-bit addressing
- Connect 555 timer to lower counter CP0
- Chain lower counter RC to upper counter CP0
- Route Q0-Q15 to SRAM A0-A15

2. Bus Transceiver Integration

- Connect 74LS645 between DIP switches and data bus
- Side A ← DIP switch lines
- Side B ← SRAM D0-D7 and LED bank
- Control DIR and /OE via SPDT switch

3. Mode Switch Implementation

- LOAD: DIR=High, /OE=Low, SRAM write enabled
- RUN: DIR=Low, /OE=Low, SRAM output enabled
- IDLE: /OE=High (optional tri-state mode)

Test Procedures

Basic Functionality Test

1. **Power-On Verification**
 - Apply 5V power
 - Verify all ICs receive clean power (check with multimeter)
 - Confirm reset brings counters to 0x0000
2. **Counter Operation**

- Set mode to RUN
- Start 555 timer or pulse manual clock
- Verify address increments: 0x0000 → 0x0001 → 0x0002...
- Check address lines with logic analyzer

3. Memory Write Test (LOAD Mode)

- Set mode to LOAD
- Set DIP switch to 0b10101010
- Press WRITE button (SRAM /WE pulse)
- Press CLOCK to increment address
- Repeat with different patterns

4. Memory Read Test (RUN Mode)

- Set mode to RUN
- Start automatic clocking
- Verify LEDs display previously stored patterns
- Confirm sequential playback

Logic Analyzer Validation Points

Channel	Signal	Purpose
0-15	Address bus	Counter operation
16-23	Data bus	Bus arbitration
24	Clock	Timing verification
25	/WE	Write pulse timing
26	/OE	Output enable timing
27	DIR	Bus direction control

Success Criteria

- ☐ Clean 16-bit address generation from 0x0000 to 0xFFFF
- ☐ No bus contention in any mode
- ☐ DIP switch data correctly stored in SRAM
- ☐ Stored patterns displayed correctly on LEDs
- ☐ Mode switching works reliably
- ☐ 555 timer provides stable clock signal

Programming Example

Create a simple LED chase pattern:

Address	Data	LED Pattern
0x0000	0b00000001	LED 0 only
0x0001	0b00000010	LED 1 only
0x0002	0b00000100	LED 2 only
0x0003	0b00001000	LED 3 only
0x0004	0b00010000	LED 4 only
0x0005	0b00100000	LED 5 only
0x0006	0b01000000	LED 6 only
0x0007	0b10000000	LED 7 only
0x0008	0b00000000	All LEDs off

Troubleshooting Guide

Issue: Counter doesn't increment

- Check 555 timer output with oscilloscope
- Verify counter power and ground connections
- Confirm clock signal reaches CP0 pins

Issue: Bus contention (dim LEDs, hot ICs)

- Verify 74LS645 DIR and /OE signals
- Check mode switch wiring
- Ensure only one device drives bus at a time

Issue: Data corruption in SRAM

- Check write pulse timing (/WE duration)
- Verify address setup/hold times
- Confirm clean power supply (add more decoupling)

Issue: Inconsistent operation

- Add RC debouncing to pushbuttons
- Check for floating inputs (add pull-ups/downs)
- Verify all bypass capacitors installed

Integration Notes for Phase 2

- Document working clock frequency and timing margins
- Note any address setup/hold time requirements
- Identify spare breadboard space for ALU addition
- Verify power supply capacity for additional ICs

Design Files

- Schematic: [Include hand-drawn or CAD schematic]
- Breadboard layout: [Photo documentation]
- Timing diagrams: [Logic analyzer captures]